

Duality

Lecture 6

October 8, 2026

Quiz

Recap From Last Time & Today's Plan

Last time...

- **Separating Hyperplane Thm $\Rightarrow$ Farkas Lemma $\Rightarrow$ Strong duality**
- **Two motivating applications + implications of duality**

Agenda for today:

- Quick Recap
- Optimality conditions and primal/dual simplex
- Complementary slackness
- Global sensitivity & Shadow prices as marginal costs
- An important application: network revenue management / pricing

Polynomially-Sized CVaR Representation

- Suppose you are choosing weights for a portfolio $x \in X$:

$$\begin{aligned} & \text{maximize}_{x \in X} f(x) \\ & \text{s.t. } \text{CVaR}(R(x)) \geq b \end{aligned}$$

- The portfolio return $R(x)$ determined by x and n different scenarios for unknown returns
- CVaR is defined as the average over the k -smallest values of $R(x)$
- (Omit dependency on x) If payoffs in the scenarios are $r_1, r_2, \dots, r_n$, the key constraint is:

$$\sum_{i=1}^k r_{[i]} \geq b, \tag{1}$$

where $r_{[1]} \leq r_{[2]} \leq \dots \leq r_{[n]}$ is the sorted vector of payoffs.

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- **Claim:**

$$\sum_{i=1}^k r_{[i]} = \min_{y \in [0,1]^n} \left\{ \sum_{i=1}^n r_i y_i : \mathbf{e}^\top \mathbf{y} = k \right\}. \tag{2}$$

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- Claim:**

$$\sum_{i=1}^k r_{[i]} = \min_{y \in [0,1]^n} \left\{ \sum_{i=1}^n r_i y_i : e^T y = k \right\}. \tag{2}$$

- By strong duality, the optimal value of LP (2) is the same as:

$$\max_{\lambda, t} \left\{ e^T \lambda + k \cdot t : \lambda + t \cdot e \leq v, \lambda \leq 0 \right\}.$$

- So (1) is satisfied if and only: $\exists \lambda, t : e^T \lambda + k \cdot t \geq b, \lambda + t \cdot e \leq v, \lambda \leq 0$.

Polynomially-Sized CVaR Representation

- Our original problem is equivalent to the optimization problem:

$$\begin{aligned} & \text{maximize}_{x, \lambda, t} f(x) \\ & \text{s.t. } x \in X \\ & \quad e^T \lambda + k \cdot t \geq b \\ & \quad \lambda + t \cdot e \leq v \\ & \quad \lambda \leq 0. \end{aligned}$$

Application in Robust Optimization

- Consider an optimization problem with an uncertain constraint:

$$\begin{aligned} & \underset{x}{\text{minimize}} \quad f(x) \\ & \text{s.t.} \quad a^T x \leq b, \quad \forall a \in \mathcal{A} := \{a \in \mathbb{R}^n : Ca \leq d\} \end{aligned} \tag{3}$$

- $\mathcal{A}$ is polyhedral set; infinitely many constraints : “semi-infinite” LP.

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Implications

Theorem (**Strong Duality**)

If $(\mathcal{P})$ has an optimal solution, so does $(\mathcal{D})$, and their optimal values are equal.

Strong duality leaves only a few possibilities for a primal-dual pair:

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Strong duality leaves only a few possibilities for a primal-dual pair:

		Dual		
		Finite Optimum	Unbounded	Infeasible
Primal	Finite Optimum	Possible	Impossible	Impossible
	Unbounded	Impossible	Impossible	Possible
	Infeasible	Impossible	Possible	?

Strong Duality and Theorems of Alternative

- Strong duality allows you to **prove** various “theorems of alternative”

Example (Farkas Lemma)

Prove that exactly one of the following is true:

- (i) $\exists x \geq 0$ such that $Ax = b$,
- (ii) $\exists \lambda$ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

Optimality for Standard-Form LPs

$$(\mathcal{P}) \min c^T x$$

$$Ax = b, \quad x \geq 0$$

$$(\mathcal{D}) \max \lambda^T b$$

$$\lambda^T A \leq c^T$$

- $(\mathcal{P})$ achieves optimality at a **basic feasible solution** x :

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 - If $B \subseteq \{1, \dots, n\}$ is a basis, the b.f.s. is: $x = [x_B, 0]$, $x_B = A_B^{-1}b$.
 - Simplex algorithm: feasibility and optimality for $(\mathcal{P})$ are given by:

$$\text{Feasibility-}(\mathcal{P}) : \quad x_B := A_B^{-1}b \geq 0 \quad (4a)$$

$$\text{Optimality-}(\mathcal{P}) : \quad c^T - c_B^T A_B^{-1} A \geq 0 \quad (4b)$$

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- $(\mathcal{D})$: same basis B can also be used to determine a **dual vector** λ :

$$\lambda^T A_i = c_i, \forall i \in B \Rightarrow \lambda^T = c_B^T A_B^{-1}, \forall i \in B.$$

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- The dual objective value corresponding to λ is: $\lambda^T b = c_B^T A_B^{-1} b = c^T x$
- λ is feasible in the dual if and only if:

$$\text{Feasibility-}(\mathcal{D}) : \quad c^T - \lambda^T A \geq 0 \quad \Leftrightarrow \quad c^T - c_B^T A_B^{-1} A \geq 0 \quad (5)$$

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Primal optimality $\Leftrightarrow$ Dual feasibility

Simplex terminates when finding a dual-feasible solution!

Solve $(\mathcal{P})$ or $(\mathcal{D})$?

$$(\mathcal{P}) \quad \min c^T x$$

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Primal simplex

- maintain a **basic feasible solution**
- basis $B \subset \{1, \dots, n\}$
- stopping criterion: dual feasibility

Dual simplex

- maintain a dual feasible solution
- stopping criterion: primal feasibility
- different from primal simplex: works with an LP with inequalities

- How to choose $(\mathcal{P})$ or $(\mathcal{D})$?
- Suppose we have x^* , λ^* and must now solve a **larger** problem, i.e., with extra decisions or extra constraints.
- *Any preference between primal and dual simplex?*
- Modern solvers include primal and dual simplex and allow concurrent runs

Optimality Conditions and Complementary Slackness

Primal-Dual Pair of Problems

$$\begin{array}{llll} (\mathcal{P}) & \underset{x}{\text{minimize}} & c^T x & \\ & & Ax \leq b & \\ & & x \geq 0 & \\ \text{variables} & & x \in \mathbb{R}^n & \end{array}$$

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Consider $x \in P, \lambda \in D$ (each feasible). How to check if they are **optimal**?

Optimality Conditions and Complementary Slackness

Primal-Dual Pair of Problems

$(\mathcal{P})$ minimize $c^T x$ $Ax \leq b$ $x \geq 0$ variables $x \in \mathbb{R}^n$	$(\mathcal{D})$ maximize $\lambda^T b$ $\lambda \geq 0$ $\lambda^T A \leq c^T$ variables $\lambda \in \mathbb{R}^m$
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Theorem (Complementary Slackness)

$x \in P$ and $\lambda \in D$ are **optimal** solutions for $(\mathcal{P})$ and $(\mathcal{D})$, respectively, **if and only if**:

$$\lambda_i (a_i^T x - b_i) = 0, i = 1, \dots, m$$

$$(\lambda^T A_j - c_j) x_j = 0, j = 1, \dots, n.$$

- Follows from primal/dual feasibility and $c^T x = b^T \lambda$

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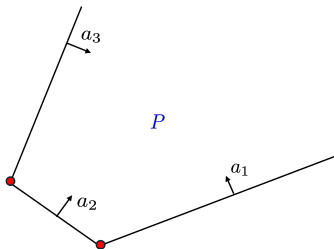
- Follows from primal/dual feasibility and $c^T x = b^T \lambda$
- Interesting insight: **non-binding constraint** $\Rightarrow$ dual variable is **zero**

Representation of Polyhedra

Important consequence of duality: alternative representation of all polyhedra

Definition

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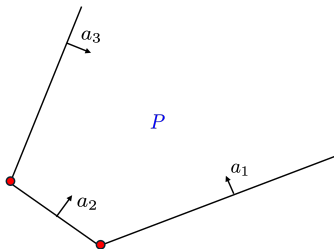
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Consider a nonempty polyhedron $P = \{x \in \mathbb{R}^n : Ax \geq b\}$. Then:

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2. Any $d \in \mathcal{C}$ with $d \neq 0$ is called a **ray** of P .



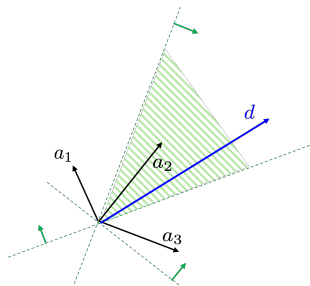
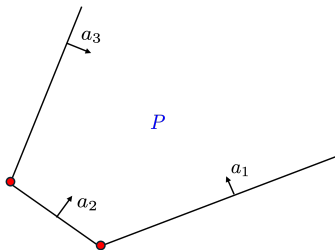
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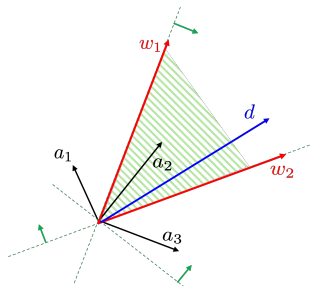
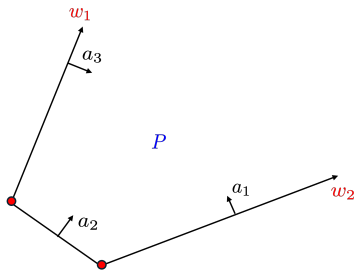
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3. Any ray d that satisfies $a_i^\top d = 0$ for $n - 1$ linearly independent a_i is called an **extreme ray** of P .



Representation of Polyhedra

Theorem (Resolution Theorem)

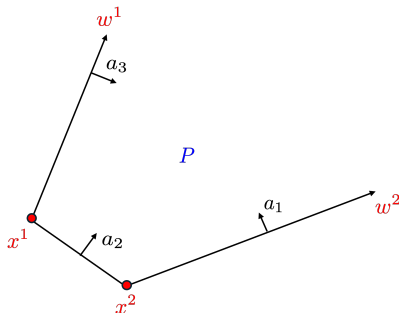
Let $P = \{x \in \mathbb{R}^n : Ax \geq b\}$ be a non-empty polyhedron, $x^1, x^2, \dots, x^k$ be its **extreme points**, and $w^1, w^2, \dots, w^r$ be its **extreme rays**. Then,

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$$P = \text{conv}(\{x^1, \dots, x^k\}) + \text{cone}(\{w^1, \dots, w^r\})$$
$$= \left\{ \sum_{i=1}^k \mu_i x^i + \sum_{j=1}^r \theta_j w^j : \mu \geq 0, e^T \mu = 1, \theta \geq 0 \right\}.$$

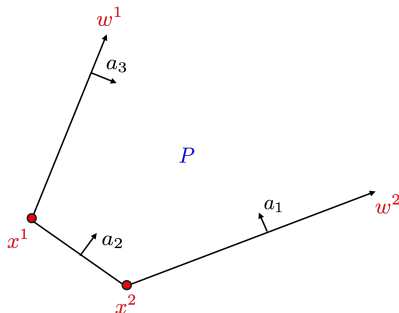


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Note: It is **not** “easy” (i.e., poly-time) to switch between these representations

Dual Variables **As Marginal Costs**

$$\begin{array}{ll} (\mathcal{P}) \min c^T x & (\mathcal{D}) \max \lambda^T b \\ Ax = b, \ x \geq 0 & \lambda^T A \leq c^T \end{array}$$

- Solved the LP and obtained x^* and λ^*
- Want to show that λ^* is the **gradient of the optimal cost with respect to b** “almost everywhere”
- Related to **sensitivity analysis**
How do the optimal value and solution depend on problem data A, b, c ?

Global Dependency On b, c

$$\begin{array}{ll} (\mathcal{P}) \min c^T x & (\mathcal{D}) \max \lambda^T b \\ Ax = b, \ x \geq 0 & \lambda^T A \leq c^T \end{array}$$

- What to show that the **optimal value** (when finite) **as a function of b** is
- What to show that the **optimal value** (when finite) **as a function of c** is

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- What to show that the **optimal value** (when finite) **as a function of b** is piecewise linear and **convex**
- What to show that the **optimal value** (when finite) **as a function of c** is piecewise linear and **concave**

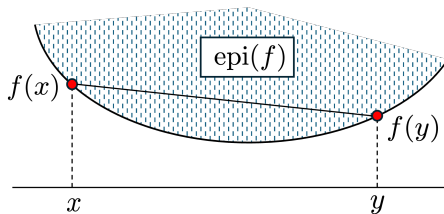
Convex and Concave Functions

Definition

$f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if X is a convex set and

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad \forall x, y \in X \text{ and } \lambda \in [0, 1]. \quad (6)$$

A function is **concave** if $-f$ is convex.



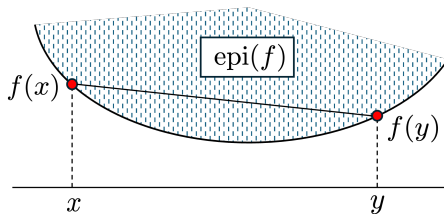
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Equivalent definition in terms of **epigraph**:

$$\text{epi}(f) = \{(x, t) \in X \times \mathbb{R} : t \geq f(x)\} \quad (7)$$

f is **convex** if and only if $\text{epi}(f)$ is a **convex** set.

Global Dependency On b

- Let $P(b) := \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}$ denote the feasible set of the primal
- Let $S := \{b \in \mathbb{R}^m : P(b) \neq \emptyset\}$: right-hand-side values that yield a feasible primal
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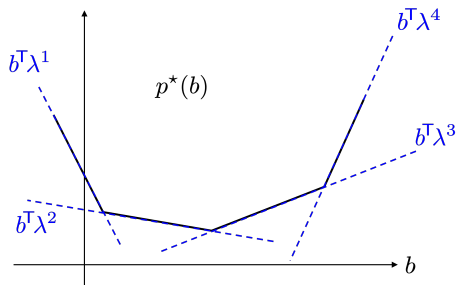
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- If $\lambda^1, \lambda^2, \dots, \lambda^r$ are the extreme points of D , then: $p^*(b) = \max_{i=1, \dots, r} b^T \lambda^i, \forall b \in S$

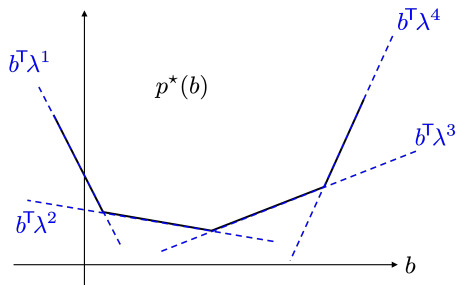


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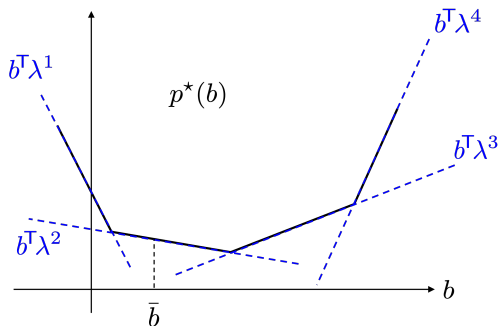
- Strong duality: $p^*(b) = \min\{c^T x : Ax = b, x \geq 0\} = \max\{\lambda^T b : \lambda^T A \leq c^T\}$
- If $\lambda^1, \lambda^2, \dots, \lambda^r$ are the extreme points of D , then: $p^*(b) = \max_{i=1, \dots, r} b^T \lambda^i, \forall b \in S$



How to prove $p^(b)$ convex?*

Global Dependency On b - Implications

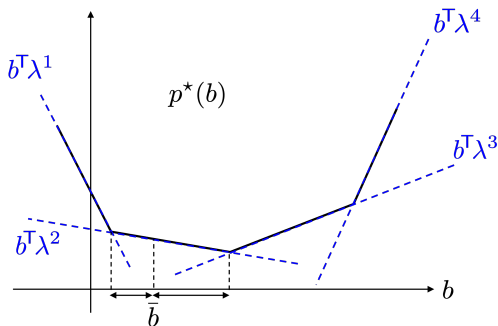
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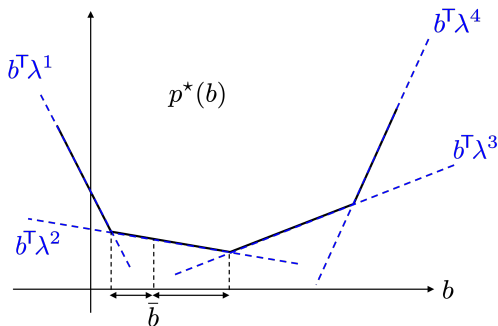
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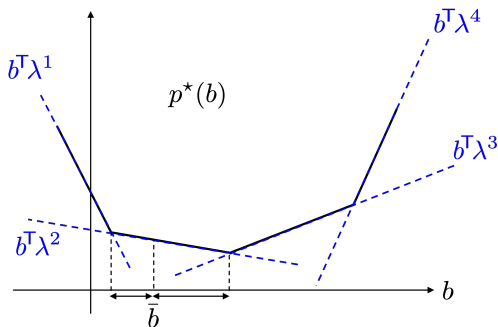


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- Modern solvers give **direct access to λ_i^* and the range**

Gurobipy: for constraint c , the attribute $c.Pi$ is λ_i^* and the range is from $c.SARHSLow$ to $c.SARHSUp$

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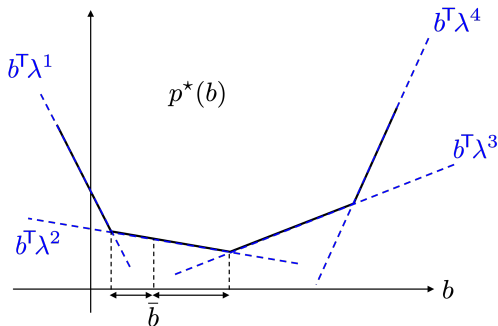
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Definition (Subgradient.)

$f : S \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ convex function. A vector $g \in \mathbb{R}^n$ is a **subgradient** of f at $\bar{x} \in S$ if

$$f(x) \geq f(\bar{x}) + g^T(x - \bar{x}), \quad \forall x \in S.$$

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- The optimal primal solution x^* **is a shadow price for the dual constraints**
- x^* remains optimal for a range of change in each objective coefficient c_j
- Modern solvers also allow obtaining the range directly
Gurobi: attributes **SAObjLow** and **SAObjUp** for each decision variable

Signs of Dual Variables Revisited

- There is a direct connection between:
 - the **optimization problem** (max/min)
 - the **constraint type** ($\leq$, $\geq$)
 - the **signs of the shadow prices**
- Given two of these, can figure out the third one!
- *What is the sign of the shadow price for a ...*
 - $\leq$ constraint in a **minimization** problem ?
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Resource matrix A :	Flight leg 1	1	0	...	1
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- Goal: decide how many itineraries of each type to sell to maximize revenue

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- **Bid-price heuristic** in network revenue management
- Broader principle of how to price “products” through resource usage/cost